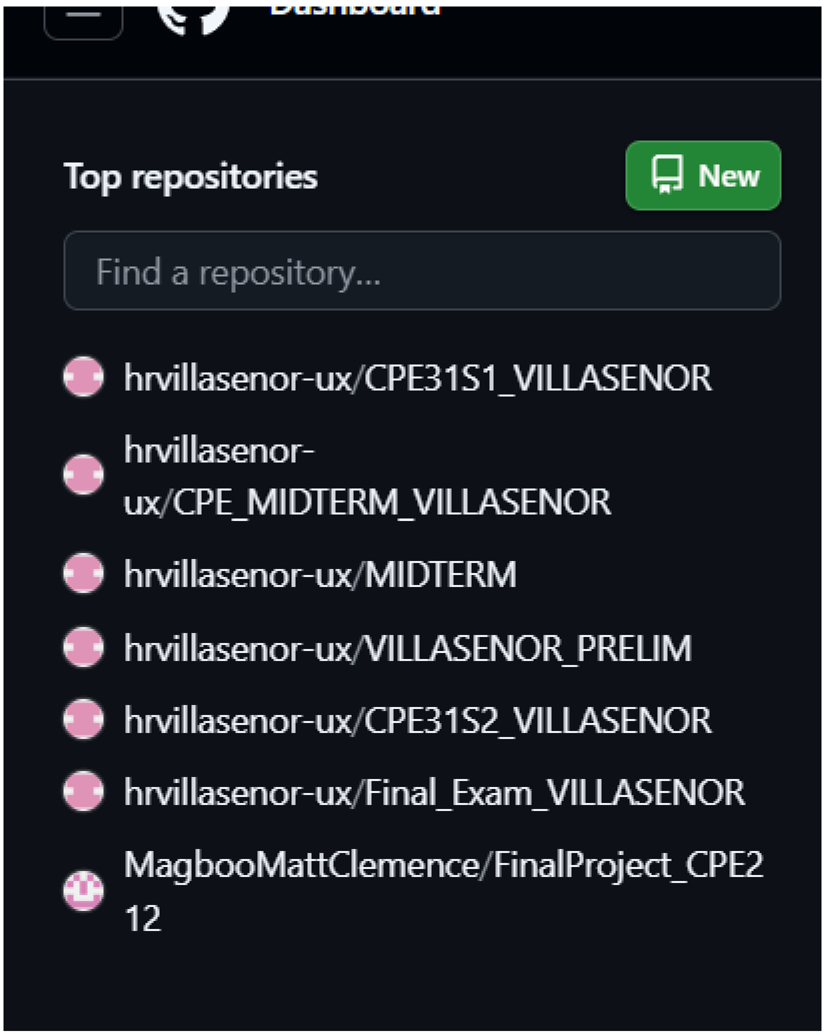
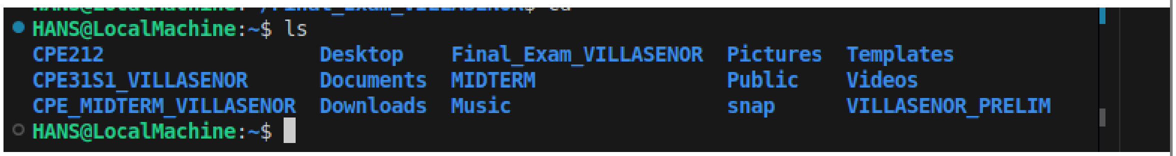
Tools Needed:
1. VM with Ubuntu, CentOS and Ansible installed 2. Web browser
Procedure:
1. Create a repository and label it as "Final_Exam_Surname" 

2. Clone your new repository in your VM



3. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.

3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers

3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
HANS@LocalMachine:~/Final_Exam_VILLASENOR$ ansible-playbook -i inventory.ini --ask-become-pass con
fig.yaml
BECOME password:

PLAY [all] ***********************************************************************************

TASK [Gathering Facts] ***********************************************************************
fatal: [192.168.56.101]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host
via ssh: HANS@192.168.56.101: Permission denied (publickey,password).", "unreachable": true}
fatal: [192.168.56.103]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host
via ssh: ssh: connect to host 192.168.56.103 port 22: No route to host", "unreachable": true}
ok: [192.168.56.104]
ok: [192.168.56.102]

TASK [motd player] ***************************************************************************
ok: [192.168.56.102]
ok: [192.168.56.104]

PLAY [all] ***********************************************************************************

TASK [Gathering Facts] ***********************************************************************
ok: [192.168.56.104]
ok: [192.168.56.102]

TASK [prometheus : Install Prometheus on Ubuntu] *********************************************
skipping: [192.168.56.104]
ok: [192.168.56.102]

TASK [prometheus : Create Prometheus user on CentOS] *****************************************
skipping: [192.168.56.102]
ok: [192.168.56.104]
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
HANS@LocalMachine:~/Final_Exam_VILLASENOR$ ansible-playbook -i inventory.ini --ask-become-pass con
fig.yaml
ok: [192.168.56.104]

TASK [prometheus : Download Prometheus tarball on CentOS] ************************************
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [prometheus : Extract Prometheus on CentOS] *********************************************
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [prometheus : Move Prometheus binary to /usr/local/bin] *********************************
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [prometheus : Create Prometheus config file] ********************************************
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [prometheus : Create Prometheus systemd service] ****************************************
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [prometheus : Start Prometheus service] *************************************************
skipping: [192.168.56.102]
changed: [192.168.56.104]
```

```
HANS@LocalMachine:~/Final_Exam_VILLASENOR$ ansible-playbook -i inventory.ini --ask-become-pass con
fig.yaml

TASK [apache : install apache and php for Ubuntu servers] ****************************************
skipping: [192.168.56.104]
ok: [192.168.56.102]

TASK [apache : install apache and php for CentOS Servers] ****************************************
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [apache : start httpd (CentOS)] *************************************************************
skipping: [192.168.56.102]
ok: [192.168.56.104]

PLAY RECAP ***************************************************************************************
192.168.56.101             : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescue
d=0    ignored=0
192.168.56.102             : ok=5    changed=0    unreachable=0    failed=0    skipped=9    rescue
d=0    ignored=0
192.168.56.103             : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescue
d=0    ignored=0
192.168.56.104             : ok=12   changed=1    unreachable=0    failed=0    skipped=2    rescue
d=0    ignored=0
```

4.4 Change Motd as "Ansible Managed by <username>"

```
HANS@LocalMachine:~/Final_Exam_VILLASENOR$ ssh HANS@192.168.56.102
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-33-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
4 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***
Ansible Managed by Hans Villasenor
Last login: Thu Nov 20 09:33:51 2025 from 192.168.56.101
HANS@Server1:~$ cat /etc/motd
Ansible Managed by Hans Villasenor
```

Go to Line/Column

VILLASENOR (17 hours ago) Ln 15, Col 17 Spaces: 2 UTF-8 LF { } YAML

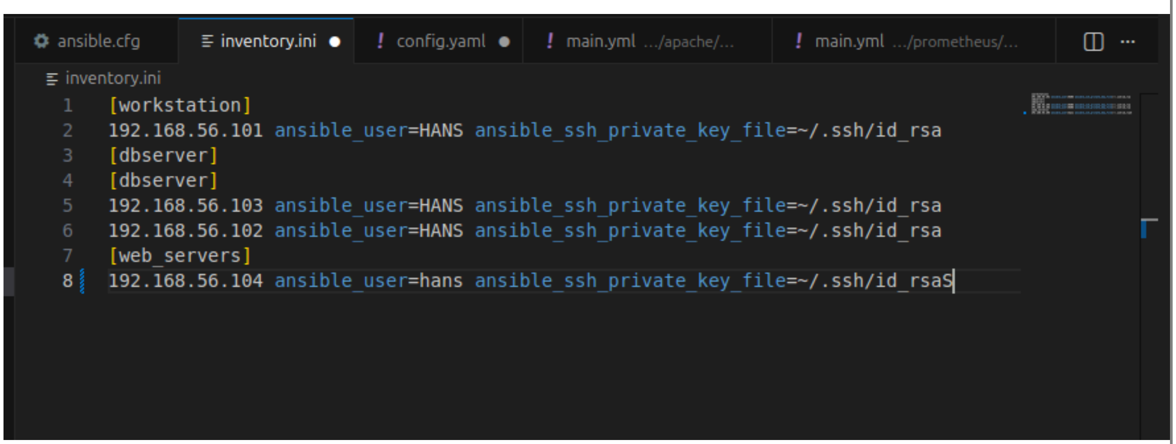
```
HANS@LocalMachine:~/Final_Exam_VILLASENOR$ ssh hans@192.168.56.104
Ansible Managed by Hans Villasenor
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Nov 20 17:34:24 2025 from 192.168.56.101
[hans@localhost ~]$ cat /etc/motd
Ansible Managed by Hans Villasenor
[hans@localhost ~]$
```

Select Indentation

4. Push and commit your files in GitHub

5. Make sure to show evidence of input (codes) process (codes successfully running) and output (evidence of installation)



config.yaml

```yaml
---
  - hosts: all
    become: true
    pre_tasks:

    - name: motd player
      copy:
          content: "Ansible Managed by Hans Villasenor\n"
          dest: /etc/motd

  - hosts: all
    become: true
    roles:
        - prometheus
        - apache
```

roles > apache > tasks > main.yml

```yaml
---

  - name: install apache and php for Ubuntu servers
    tags: apache2-ubuntu
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache and php for CentOS Servers
    tags: centos-httpd
    dnf:
      name:
        - httpd
        - php
      state: latest
    when: ansible_os_family == "RedHat"

  - name: start httpd (CentOS)
    tags: centos-httpd
    service:
      name: httpd
      state: started
      enabled: true
    when: ansible_os_family == "RedHat"
```

roles > prometheus > tasks > ! main.yml

```yaml
---

- name: Install Prometheus on Ubuntu
  tags: prometheus
  apt:
    name: prometheus
    state: latest
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: Create Prometheus user on CentOS
  tags: prometheus
  user:
    name: prometheus
    shell: /sbin/nologin
  when: ansible_os_family == "RedHat"

- name: Download Prometheus tarball on CentOS
  tags: prometheus
  get_url:
    url: https://github.com/prometheus/prometheus/releases/download/v2.45.0/promet
    dest: /tmp/prometheus.tar.gz
  when: ansible_os_family == "RedHat"

- name: Extract Prometheus on CentOS
  tags: prometheus
  unarchive:
    src: /tmp/prometheus.tar.gz
    dest: /opt/
    remote_src: yes
  when: ansible_os_family == "RedHat"
```

ansible.cfg | inventory.ini ● | ! config.yaml | ! main.yml .../apache/... | ! main.yml .../prometheus/... ✕

roles > prometheus > tasks > ! main.yml

```yaml

- name: Move Prometheus binary to /usr/local/bin
  tags: prometheus
  copy:
    remote_src: yes
    src: /opt/prometheus-2.45.0.linux-amd64/prometheus
    dest: /usr/local/bin/prometheus
    mode: '0755'
  when: ansible_os_family == "RedHat"

- name: Create Prometheus config file
  tags: prometheus
  copy:
    dest: /etc/prometheus.yml
    content: |
      global:
        scrape_interval: 15s

      scrape_configs:
        - job_name: 'node_exporter'
          static_configs:
            - targets: ['192.168.56.101:9100', '192.168.56.104:9100']
  when: ansible_os_family == "RedHat"
```

```yaml
- name: Create Prometheus systemd service
  tags: prometheus
  copy:
    dest: /etc/systemd/system/prometheus.service
    content: |
      [Unit]
      Description=Prometheus Monitoring
      After=network.target

      [Service]
      User=prometheus
      ExecStart=/usr/local/bin/prometheus --config.file=/etc/prometheus.yml

      [Install]
      WantedBy=multi-user.target
  when: ansible_os_family == "RedHat"

- name: Start Prometheus service
  tags: prometheus
  systemd:
    name: prometheus
    enabled: true
    state: started
  when: ansible_os_family == "RedHat"
```

```
Last login: Thu Nov 20 17:34:24 2025 from 192.168.56.101
[hans@localhost ~]$ cat /etc/motd
Ansible Managed by Hans Villasenor
[hans@localhost ~]$ which prometheus
/usr/local/bin/prometheus
[hans@localhost ~]$
```

```
HANS@Server1:~$ which /etc/prometheus
HANS@Server1:~$ dpkg -l | grep prometheus
ii  prometheus                               2.45.3+ds-2ubuntu0.3                    amd64
      monitoring system and time series database
ii  prometheus-node-exporter                 1.7.0-1ubuntu0.3                        amd64
      Prometheus exporter for machine metrics
ii  prometheus-node-exporter-collectors      0.0~git20231018.f5c56e7-1               all
      Supplemental textfile collector scripts for Prometheus node_exporter
ii  python3-prometheus-client                0.19.0+ds1-1                            all
      Python 3 client for the Prometheus monitoring system
HANS@Server1:~$
```

VILLASENOR (18 hours ago)   Ln 8, Col 22   Spaces: 2   UTF-8   LF   { } YAML

5.1 For your final exam to be counted, please paste your repository link as an answer in this exam.

https://github.com/hrvillasenor-ux/Final_Exam_VILLASENOR.git

Note: Extra points if you will implement the said services via containerization.